Exact estimator for plasmid conjugation rates

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ABSTRACT Accurate estimates of plasmid conjugation rates are required for quantitative studies of horizontal gene transfer. Kosterlitz and colleagues adapted the Luria–Delbrück fluctuation assay to conjugation. Their estimator uses the fraction of replicate cultures containing no transconjugants. This zero-class statistic is robust, but it does not use the numbers of transconjugants in positive cultures. Here, the same stochastic conjugation model is used to derive the complete transconjugant-count distribution. Primary donor-to-recipient transfer events form a time-dependent Poisson process. Each event starts a transconjugant lineage that grows by cell division and secondary transfer. Under the assumptions of the published model, the resulting count has a compound-Poisson probability-generating function. Its logarithmic coefficients are obtained by one-dimensional numerical integration, after which a general coefficient recursion gives the complete probability mass function. Bernoulli sampling incorporates the plated fraction and colony-establishment probability. The resulting likelihood uses every colony count and, under complete detection, reduces to the published zero-class method when counts are recorded only as absent or present. Empirical probability-generating-function fitting and Fourier inversion provide independent computational methods and permit extensions for which the coefficient recursion is not convenient. These results combine the conjugation model of Kosterlitz and colleagues with full-count methods developed for fluctuation analysis by Gerrish and by Vázquez-Mendoza and colleagues.

KEYWORDS conjugation; fluctuation analysis; horizontal gene transfer; Luria–Delbrück distribution; plasmid; probability-generating function

Introduction

Horizontal gene transfer contributes to bacterial adaptation by moving genes among cells that are not related by direct descent (Thomas and Nielsen 2005; Norman *et al.* 2009; Redondo-Salvo *et al.* 2020). Conjugative plasmids are important agents of this process. They carry genes for their own transfer and may also carry traits that affect bacterial survival, including antimicrobial resistance (San Millan 2018; Antimicrobial Resistance Collaborators 2022). Quantitative studies of plasmid ecology therefore require an estimate of the conjugation rate under a specified set of donor, recipient, plasmid, and environmental conditions.

The standard experimental design mixes plasmid-bearing donors with plasmid-free recipients and measures donor, recipient, and transconjugant densities after a period of growth. Classical estimators are based on deterministic mass-action models (Levin *et al.* 1979; Simonsen *et al.* 1990). Their accuracy can be affected by unequal growth rates, unequal transfer rates from donors and transconjugants, resource limitation, and low transconjugant numbers (Huisman *et al.* 2022; Kosterlitz *et al.* 2022). These problems are most severe when the first few transfer events have a large effect on the final transconjugant population.

Kosterlitz et al. (2022) addressed this problem by treating transconjugant production as a stochastic process. Their Luria–Delbrück method estimates the primary donor conjugation rate from the fraction of independent cultures that contain no transconjugants. The method follows the zero-class argument of Luria and Delbrück (1943): the probability of observing no event depends on the integrated rate at which the first event occurs, but not on the later expansion of the resulting lineage. This property allows the primary conjugation rate to be estimated even when transconjugants and donors have different transfer rates.

The zero class is useful because it is simple and relatively insensitive to the distribution among positive counts. It also discards information. A culture with 1 transconjugant and a culture with 1,000 transconjugants have the same binary score.

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In mutation-rate estimation, the complete Luria–Delbrück distribution has long been used to distinguish late events from early events that produce large clones (Lea and Coulson 1949; Zheng 1999). Methods for computing this distribution include coefficient recursions based on the logarithm of its probability-generating function (Ma *et al.* 1991; Sarkar *et al.* 1992; Zheng 2005).

The treatment of sampling is also relevant. Only a fraction of a culture is usually plated, and only a fraction of plated cells may form colonies (Stewart 1991; Jones *et al.* 1994; Zheng 2008). Gerrish (2008) composed the Luria–Delbrück probability-generating function with the probability-generating function of Bernoulli sampling and derived an efficient recursion for the resulting counts. This calculation made full likelihood analysis compatible with the extended Jones protocol, in which cultures are grown to a large size and a known fraction is plated.

More recently, Vázquez-Mendoza *et al.* (2024) considered a fluctuation model whose probability-generating function did not lead directly to a convenient closed-form probability mass function. They fitted the theoretical function to the empirical probability-generating function and also recovered count probabilities by Fourier inversion. These methods separate the biological derivation of a probability-generating function from the numerical calculation used for inference (Nakamura and Pérez-Abreu 1993; Lange 1982).

Here these developments are applied to the stochastic conjugation model of Kosterlitz *et al.* (2022). The result is a full-count extension of their method. The derivation also shows that the general coefficient recursion remains available: closed-form coefficients are not required because they can be evaluated by numerical integration.

The conjugation process

Consider a well-mixed culture of volume V. Let $D(s)$ and $R(s)$ be donor and recipient densities at time s. As in the basic model of Kosterlitz *et al.* (2022), assume deterministic exponential growth during the assay,

$$D(s) = D_0 e^{\psi_D s}, \qquad R(s) = R_0 e^{\psi_R s}, \tag{1}$$

where ψ_D and ψ_R are per-capita growth rates. Recipient loss through transfer is neglected. The primary transfer rate from donors to recipients, expressed as a total rate for the culture, is

$$a(s) = V\gamma_D D(s) R(s), \tag{2}$$

where γ_D is the donor conjugation-rate constant.

Let $T(s)$ be the number of transconjugants. Each transconjugant produces 1 additional transconjugant either by cell division or by transfer to a recipient. Its per-capita birth rate is therefore

$$b(s) = \psi_T + \gamma_T R(s), \tag{3}$$

where ψ_T is the transconjugant growth rate and γ_T is the secondary conjugation-rate constant. In an interval $\mathrm{d}s$, the transition rate from T to $T+1$ is

$$a(s) + b(s)T. \tag{4}$$

Thus, primary transfer is time-dependent immigration and transconjugant increase is a time-dependent linear birth process.

The complete transconjugant distribution

Suppose that a primary transfer occurs at time s and starts a lineage with 1 transconjugant. Define

$$B(s,t) = \int_s^t b(u)\,\mathrm{d}u, \qquad q_s = e^{-B(s,t)}. \tag{5}$$

The size $X_{s,t}$ of this lineage at assay time t has the geometric distribution

$$\Pr(X_{s,t} = k) = q_s(1-q_s)^{k-1}, \qquad k \geq 1, \tag{6}$$

with probability-generating function

$$g_{s,t}(z) = \frac{q_s z}{1-(1-q_s)z}. \tag{7}$$

For the exponential recipient trajectory in Equation (1),

$$B(s,t) = \psi_T(t-s) + \frac{\gamma_T R_0}{\psi_R}\left(e^{\psi_R t} - e^{\psi_R s}\right), \tag{8}$$

with the continuous limit $\gamma_T R_0(t-s)$ for the second term when $\psi_R = 0$.

Primary transfers form an inhomogeneous Poisson process with rate $a(s)$. Conditional on their initiation times, their descendant lineages are independent. The probability-generating function of the total transconjugant count is therefore

$$G_t(z) = \exp\left\{\int_0^t a(s)\left[g_{s,t}(z) - 1\right]\mathrm{d}s\right\}. \tag{9}$$

This is a time-dependent compound-Poisson distribution. A related closed-form probability-generating function was considered in the supporting analysis of Kosterlitz *et al.* (2022) for a restricted case, using results of Keller and Antal (2015). Equation (9) retains secondary conjugation through the time-dependent birth rate in Equation (3).

Define

$$A(t) = \int_0^t a(s)\,\mathrm{d}s, \tag{10}$$

$$\lambda_k(t) = \int_0^t a(s) q_s (1-q_s)^{k-1}\,\mathrm{d}s, \qquad k \geq 1. \tag{11}$$

The value λ_k is the expected number of primary transfers whose lineages contain exactly k cells at time t. Because $\sum_{k\geq 1}\lambda_k = A$, Equation (9) can be written

$$\log G_t(z) = -A + \sum_{k=1}^{\infty} \lambda_k z^k. \tag{12}$$

Let $p_n = \Pr[T(t) = n]$. Differentiating Equation (12) and equating coefficients gives the general recursion (Ma *et al.* 1991; Sarkar *et al.* 1992)

$$p_0 = e^{-A}, \qquad p_n = \frac{1}{n}\sum_{k=1}^{n} k\lambda_k p_{n-k}. \tag{13}$$

The λ_k need not have closed forms. They can be calculated by one-dimensional quadrature. Probabilities through an observed maximum $n_{\max}$ require only $\lambda_1, \ldots, \lambda_{n_{\max}}$.

The zero class follows immediately from Equation (10). Under Equation (1),

$$p_0 = \exp\left[-V\gamma_D D_0 R_0 \frac{e^{(\psi_D+\psi_R)t} - 1}{\psi_D + \psi_R}\right], \tag{14}$$

which is the estimator-defining probability in Kosterlitz *et al.* (2022). The published method is therefore the $n = 0$ part of the complete distribution.

Plating and colony establishment

Let ρ be the effective probability that a transconjugant present at time t produces a counted colony. It includes the plated fraction and colony-establishment probability. Under independent sampling,

$$C \mid T \sim \text{Binomial}(T, \rho), \tag{15}$$

and the probability-generating function of the colony count C is

$$H_t(z) = G_t(1 - \rho + \rho z). \tag{16}$$

This is the same probability-generating-function composition used to represent dilution and plating in fluctuation assays (Stewart 1991; Jones *et al.* 1994; Gerrish 2008; Zheng 2008; Vázquez-Mendoza *et al.* 2024).

The composition can again be put in a form suitable for Equation (13). Set

$$d_s = q_s + (1 - q_s)\rho. \tag{17}$$

A lineage initiated at s contributes no observed colony with probability

$$\Pr(Y_s = 0) = \frac{q_s(1-\rho)}{d_s}, \tag{18}$$

and contributes $j \geq 1$ colonies with probability

$$\Pr(Y_s = j) = \frac{q_s \rho}{d_s^2} \left[\frac{(1-q_s)\rho}{d_s} \right]^{j-1}. \tag{19}$$

Define

$$\widetilde{\lambda}_j = \int_0^t a(s) \frac{q_s \rho}{d_s^2} \left[\frac{(1-q_s)\rho}{d_s} \right]^{j-1} \mathrm{d}s, \tag{20}$$

$$\widetilde{A} = \int_0^t a(s) \frac{\rho}{d_s} \, \mathrm{d}s. \tag{21}$$

Then $\sum_{j \geq 1} \widetilde{\lambda}_j = \widetilde{A}$, and the observed-count probabilities $h_n = \Pr(C = n)$ satisfy

$$h_0 = e^{-\widetilde{A}}, \qquad h_n = \frac{1}{n} \sum_{j=1}^{n} j \widetilde{\lambda}_j h_{n-j}. \tag{22}$$

When $\rho < 1$, an observed zero can result either from no primary transfer or from failure to sample the descendants of a transfer. This distinction must be included if a plated zero class is used. Equation (14) is recovered when $\rho = 1$ and transconjugant lineages do not become extinct.

Inference from colony counts

For independent cultures with observed counts $c_1, \ldots, c_m$, the full log likelihood is

$$\ell(\theta) = \sum_{i=1}^{m} \log h_{c_i}(\theta), \tag{23}$$

where θ contains γ_D and any parameters not fixed by separate measurements. If D, R, ψ_T, γ_T, and ρ are fixed, each coefficient in Equation (20) is proportional to γ_D. The required integrals can then be calculated once apart from this common factor, and one-dimensional likelihood maximization is fast.

Equation (23) uses the complete set of observed counts. The binary likelihood used by the zero-class method is obtained by replacing each c_i with $I(c_i > 0)$. Because this replacement is a deterministic reduction of the data, the count experiment cannot contain less Fisher information about a correctly specified parameter than its binary reduction. The gain may be small when nearly every positive culture contains a single detected lineage. It can be substantial when the positive counts vary in ways that distinguish the number and timing of transfer events. As in conventional fluctuation analysis, this gain is accompanied by greater sensitivity to assumptions about lineage growth and sampling (Gerrish 2008; Ycart 2013).

Empirical probability-generating-function estimation

The empirical probability-generating function is

$$\widehat{H}_m(z) = \frac{1}{m} \sum_{i=1}^{m} z^{c_i}. \tag{24}$$

Following Vázquez-Mendoza *et al.* (2024), parameters may be estimated by minimizing the distance between Equation (24) and the theoretical function in Equation (16) at several real values of z between 0 and 1. The original zero class is retained because $\widehat{H}_m(0)$ is the observed fraction of zero counts. Values $z > 0$ add information from positive cultures while limiting the effect of very large counts.

For real evaluation points z_j and z_k, the model covariance is

$$\operatorname{Cov}\left[\widehat{H}_m(z_j), \widehat{H}_m(z_k)\right] = \frac{H_t(z_j z_k) - H_t(z_j) H_t(z_k)}{m}. \tag{25}$$

This covariance permits weighted minimum-distance estimation. The evaluation points should be selected by simulation under the intended experimental range rather than copied from another assay. Confidence intervals can be obtained by resampling whole cultures or by parametric simulation.

Fourier inversion

Count probabilities may also be obtained by evaluating the probability-generating function on a complex circle and applying a discrete inverse Fourier transform (Lange 1982; Vázquez-Mendoza *et al.* 2024). A radially scaled form is

$$h_n \simeq \frac{1}{M r^n} \sum_{\ell=0}^{M-1} H_t\left(r e^{2\pi i \ell / M}\right) e^{-2\pi i n \ell / M}, \qquad 0 < r < 1. \tag{26}$$

The choice $r < 1$ reduces wrap-around error from the infinite right tail. Values of M and r should be varied until all likelihood contributions for the observed counts are stable. Under the present model, Equation (22) is simpler. Fourier inversion remains useful as an independent numerical check and for extensions in which clone probabilities are obtained numerically.

Experimental design and model checks

The binary selective outgrowth used by Kosterlitz *et al.* (2022) should be replaced by direct count measurement. Independent donor–recipient cultures are stopped at a specified time, further growth and conjugation are stopped, and a known fraction of each culture is plated on transconjugant-selective medium. The numerical colony count, including zero, is recorded for every culture. Selective conditions must prevent new conjugation on the plate; this must be checked for each system because post-plating transfer can increase apparent transconjugant counts (Smit and van Elsas 1990; Kosterlitz *et al.* 2022).

The effective sampling probability ρ should be measured with transconjugant controls. It may differ from the volume fraction because of cell death, incomplete colony establishment, or cell clumping. If several disjoint aliquots are plated, their counts can be summed and the total plated fraction used. If a dilution is selected after its count is observed, the selection rule should be included in the likelihood; a prespecified dilution or a prespecified set of plates is simpler. Counts above a plate-reading limit should be retained as right-censored observations. For a threshold L, their likelihood contribution is

$$\Pr(C \geq L) = 1 - \sum_{n=0}^{L-1} h_n. \tag{27}$$

The extended Jones principle of using a large culture followed by dilution can increase information while keeping colony counts manageable (Jones *et al.* 1994; Gerrish 2008). In a conjugation assay, assay duration cannot be increased without limit. Recipient depletion, stationary phase, density-dependent transfer, and secondary conjugation may violate Equations (1)–(4). The culture volume, stopping time, and plated fraction should therefore be selected by simulation so that counts are measurable while transconjugants remain rare enough for the Kosterlitz assumptions to hold.

The primary rate γ_D controls lineage initiation, whereas ψ_T and γ_T control lineage size. These parameters may be weakly identified from a single endpoint. Separate measurements of transconjugant growth and secondary conjugation, as used by Kosterlitz *et al.* (2022), reduce this problem. Multiple stopping times or recipient densities provide additional separation. Uncertainty in independently estimated rates and in ρ should be propagated by profile likelihood, parametric bootstrap, or joint inference.

Model checking should include the zero fraction, the empirical probability-generating function, the central count distribution, and the upper tail. Agreement of $\widehat{\gamma}_D$ across stopping times and plating fractions provides a direct check of the growth and sampling assumptions. Failure of these checks indicates that a more detailed process is required.

Discussion

The complete count theory follows from 3 existing developments. Kosterlitz et al. (2022) provided the biological process and identified the zero class that isolates primary transfer. Gerrish (2008) showed how plating can be represented by probability-generating-function composition and how full likelihood inference can remain practical after dilution. Vázquez-Mendoza et al. (2024) showed how an empirical probability-generating function or Fourier inversion can be used when direct probability calculations are difficult. Together these methods yield a count-based conjugation assay without changing the main assumptions of the published Luria–Delbrück method.

The main mathematical result is that the Kosterlitz process is compound Poisson. A primary transfer starts an independent transconjugant lineage, and the final count is the sum of the sizes of all such lineages. This structure places the logarithm of the probability-generating function in the form required by the Ma–Sandri–Sarkar recursion. The individual coefficients are numerical integrals rather than closed expressions, but the recursion itself is unchanged. An empirical probability-generating function or Fourier inversion is therefore not required for the basic model, although both remain useful.

The compound-Poisson representation also extends beyond the pure-birth case. If a lineage has probability-generating function $g_{s,t}(z)$ under a model with death, plasmid loss, or a lag period, Equation (9) remains valid as long as primary events have deterministic intensity and their lineages remain independent. Lineage extinction then contributes to the zero probability, as considered by Kosterlitz *et al.* (2022) and in stochastic establishment models (Alexander and MacLean 2020). The required logarithmic coefficients can be obtained from a numerical solution for the lineage distribution or by Fourier inversion.

The representation fails when transconjugant lineages interact strongly through recipient depletion or when donor and recipient trajectories depend materially on the realized transfer history. In that case the process is no longer a sum of independent lineages. A joint master equation or stochastic simulation can still provide a theoretical or empirical probability-generating function, to which the methods of Vázquez-Mendoza *et al.* (2024) can be applied. The count-based framework therefore provides both an exact method for the current model and a route to more detailed models.

Data availability

No empirical data were generated or analyzed in this study. The method is defined by the equations in the article.

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Conflicts of interest

The author declares no conflict of interest.

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